A FLAT FAMILY OVER AN INTEGRAL BASE WITH INTEGRAL GENERIC FIBRE IS INTEGRAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The proof reduces the geometric statement to its algebraic analogue and proves that analogue in place; the single citation supplies torsion freeness only.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- A flat module over a domain is torsion free. More on Algebra, Lemma more-algebra-lemma-flat-torsion-free, used at proof lines 1743-1746. Gives the injectivity B -> B (x)_A K.

Source reference. Original lemma 0538: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6596–6596. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to Y$ be a flat morphism of schemes. If Y is integral and the generic fibre of f is integral, then X is integral.*

Proof. The algebraic analogue is this: let A be a domain with fraction field K and let B be a flat A-algebra such that $B \otimes_A K$ is a domain. Then B is a domain. This is true because B is torsion free by More on Algebra, Lemma 0538 and hence $B \subset B \otimes_A K$. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.